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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VOICE BEYOND SILENCE: ECHOGENESIS. A LONGITUDINAL DEGRADATION-ROBUST FRAMEWORK FOR VOICE PRESERVATION IN PROGRESSIVE SPEECH LOSS.

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INTRODUCTION

A person's voice is more than just a way to communicate. It is a part of their identity and helps others recognize them. People with conditions such as ALS, Parkinson's disease, stroke, laryngectomy, or dysarthria slowly lose their ability to speak. As their voice changes, it becomes difficult for them to communicate and express themselves. Most existing voice cloning and Text-to-Speech (TTS) systems need clear and healthy voice recordings to create a good voice model. They also cannot adjust when a person's voice keeps changing because of the disease. As a result, the generated speech may not sound like the person's original voice. EchoGenesis is an AI-based voice preservation system that can learn from short or partially affected voice recordings. It preserves the person's unique voice and generates clear, natural speech, allowing them to continue communicating in a voice that sounds like their own even as their condition progresses.

ABSTRACT

EchoGenesis is a voice preservation system for people whose speech changes due to ALS, Parkinson’s disease, stroke, cancer treatment, or laryngectomy. Existing voice cloning and TTS systems mainly use healthy speech and cannot handle changes in a person’s voice over time. EchoGenesis preserves the person’s original and unique voice even when their speech becomes unclear. It uses a degradation robust speaker encoder to identify the person’s voice from affected speech recordings.It compares recent recordings with earlier, healthier recordings to maintain the original voice identity. A few shot TTS model generates natural speech using only a small number of voice recordings.Prosody aware chunked synthesis improves the clarity of longer sentences and makes the speech easier to understand.The system is evaluated using speaker similarity, WER, and a degradation robustness curve.

Keywords:
Voice Cloning; Artificial Intelligence; Deep Learning; ALS; Throat Cancer; Speech Preservation; Assistive Technology; Personalized Voice Model, Word Error Rate(WER).

BASE PAPER

AUTHOR	TITLE	METHODOLOGY	LIMITATION	DATA SET	ADDRESSES
Vinotha R, Hepsiba D, L.D. Vijay Anand, Deepak John Reji. 2024. arXiv ID: 2401.11771v2	Empowering Communication: Speech Technology for Indian and Western Accents through AI-powered Speech Synthesis	Data Collection, Pre-processing, Speaker Encoder using LSTM(long short term memory), Synthesizer using Text-to-Mel, WaveRNN Vocoder, Noise Reduction, and Evaluation.	The system still finds it difficult to accurately copy complex emotions and expressions in natural human speech.	The study used LibriSpeech, NPTEL2020 Indian English, LJSpeech, VCTK, and AccentDB datasets.	Voice cloning struggle to recreate subtle emotional cues, leaving synthesized speech feeling unnaturally flat.
Anuprabha M, Krishna Gurugubelli, Anil Kumar Vuppala. 15 August 2025. arXiv:2508.05102v3	Fairness in dysarthric speech synthesis means reducing bias in F5-TTS speech cloning.	F5-TTS clones dysarthric speech from TORGO and evaluates it for intelligibility, speaker similarity, prosody, and fairness.	Synthetic speech can be harder to understand in severe dysarthria, and too much synthetic data may reduce recognition accuracy.	The study uses the TORGO dataset with 8 dysarthric and 7 healthy speakers, covering low, medium, and high dysarthria severity.	Separating vocal decline from speaker traits allows intelligible, customized voice retention across changing stages.
Birger Moell, Fredrik Sand Aronsson. 3 March 2025. arXiv:2503.01266v1	Voice Cloning for Dysarthric Speech Synthesis: Addressing Data Scarcity in Speech-Language Pathology	Data Collection, TORGO Dataset, Audio Selection, ElevenLabs Voice Cloning, Synthetic Speech Generation, SLP(single layer perceptron) Evaluation, and Performance Evaluation.	Small number of samples, only English speakers, some samples had only single words, and some artificial speech had language mistakes that made it sound unnatural.	The study used TORGO dataset, containing speech recordings from speakers with dysarthria (ALS/CP) and healthy control.	Disentangling progressive speech impairment from identity enables clear, personal voice preservation over time.

LITERATURE SURVEY

AUTHOR	TITLE	METHODOLOGY	LIMITATION	DATA SET	ADDRESSES
Ye Jia, Yu Zhang, Ron J. Weiss, Quan Wang, Jonathan Shen, Fei Ren, Zhifeng Chen, Patrick Nguyen, Ruoming Pang, Ignacio Lopez Moreno, Yonghui Wu. 2 Jan 2019. arXiv:1806.04558v4 [cs.CL]	Transfer Learning from Speaker Verification to Multispeaker Text-To-Speech Synthesis	Three-part decoupled system peaker encoder, Tacotron 2 based synthesizer, and WaveNet vocoder.	Reduced speaker similarity on unseen speakers, inability to transfer accents, and fixed embedding capacity limits.	LibriSpeech, Voice Cloning Toolkit (VCTK), VoxCeleb, VoxCeleb2, and a proprietary voice search corpus.	Using zero-shot voice cloning for low-burden data augmentation improves Dysarthric Automatic Speech Recognition (ASR) performance and cross-corpus generalization.
Giuseppe Ruggiero, Enrico Zovato, Luigi Di Caro, Vincent Pollet. 10 Feb 2021. arXiv:2102.05630v1 [cs.SD]	Voice Cloning: A Multi-Speaker Text-to-Speech Synthesis Approach Based on Transfer Learning	Three-part decoupled system Speaker Encoder (SE), Sequence to Sequence (Seq2Seq) Tacotron 2-based synthesizer, and WaveRNN based Vocoder.	Reduced Speaker Similarity (SS) on unseen speakers, inability to capture target Speaker Prosody (SP), and lower naturalness compared to single-speaker models	LibriTTS, VoxCeleb, VoxCeleb2, CSTR Voice Cloning Toolkit (VCTK), and a proprietary voice corpus	Using zero-shot voice cloning for low-burden data augmentation improves Dysarthric Automatic Speech Recognition (ASR) performance and cross-corpus generalization.
Satwinder Singh, Qianli Wang, Zihan Zhong, Clarion Mendes, Mark Hasegawa-Johnson, Waleed Abdulla, Seyed Reza Shahamiri. 18 June 2026. arXiv:2606.19823v1	Low-Burden Data Augmentation for Dysarthric ASR via Zero-Shot Voice Cloning	The study uses zero-shot voice cloning, synthetic speech generation, Whisper fine tuning, and WER-based evaluation.	Synthetic data may create unnatural sounds, and its effectiveness can change depending on the severity of dysarthria.	The study used TORGO dataset approximately 23 hours of speech from 8 dysarthric speakers (5 male, 3 female), mainly with cerebral palsy (CP) and ALS.	Using zero-shot voice cloning for low-burden data augmentation improves dysarthric ASR performance and cross-corpus generalization.

LITERATURE SURVEY

AUTHOR	TITLE	METHODOLOGY	LIMITATION	DATA SET	ADDRESSES
Seonggyu Lee, Sangwook Han, Sein Cheong, Jong Won Shin. 9 Aug 2025. arXiv:2508.06840v1 [eess.AS]	FlowSE: Flow Matching-based Speech Enhancement	Three-part decoupled system—Speaker Encoder (SE), Sequence-to-Sequence (Seq2Seq) Tacotron 2-based synthesizer, and WaveRNN-based Vocoder.	Reduced Speaker Similarity (SS) on unseen speakers, inability to capture target Speaker Prosody (SP), and lower naturalness compared to single-speaker models.	LibriTTS, VoxCeleb, VoxCeleb2, CSTR Voice Cloning Toolkit (VCTK), and a proprietary voice corpus.	Using zero-shot voice cloning for low-burden data augmentation improves Dysarthric Automatic Speech Recognition (ASR) performance and cross-corpus generalization.
Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, Ilya Sutskever. 6 Dec 2022. arXiv:2212.04356v1 [eess.AS]	Robust Speech Recognition via Large-Scale Weak Supervision	Uses large-scale multilingual audio data, filters transcripts, converts audio into Mel-spectrograms, and trains an encoder-decoder Transformer model.	It may make mistakes when transcribing long speech and needs a lot of computer power.	The study used 680,000-hour web-scale audio-transcript dataset.	Provides accurate speech recognition in different languages and environments without needing specific training for each one.
Liang Lu, Lingpeng Kong, Chris Dyer, and Noah A. Smith. 5 Jun 2017. arXiv:1702.06378v4 [cs.CL]	Multitask Learning with CTC(connectionlist temporal classification)and Segmental CRF for Speech Recognition	Jointly trains CTC and Segmental CRF(conditional random field)using a shared LSTM-RNN encoder with an interpolated loss on the TIMIT(texas instrument institution)speech dataset.	SCRF needs a lot of computer power, and its context-based embedding may slightly reduce recognition accuracy.	The study uses the TIMIT speech recognition dataset. The experiments follow the standard protocol using the core test set of 192 expressions.	Improves speech recognition accuracy using CTC and SCRF(segmental conditional random field), while making training faster through CTC pretraining.

PROBLEM STATEMENT

Progressive speech disorders cause individuals to lose both their ability to speak and their unique vocal identity. Current voice cloning solutions generated by various researchers cannot effectively preserve or adapt a person's voice as it deteriorates over time.

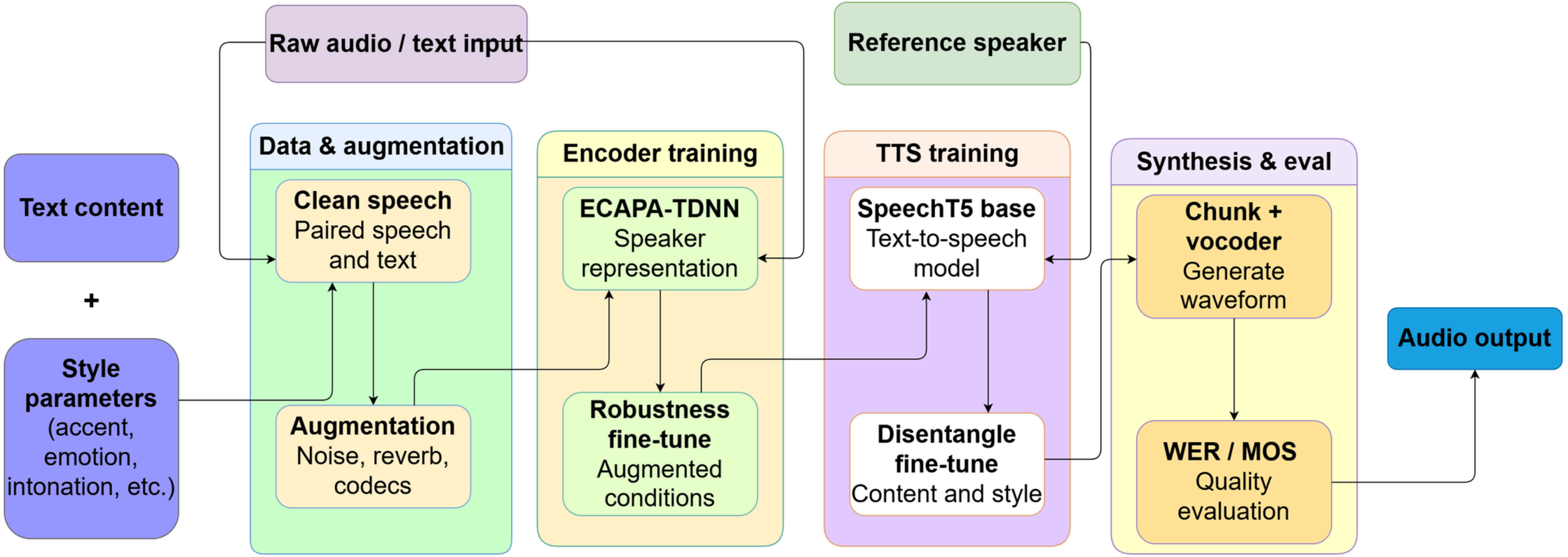
OBJECTIVE

- Design and develop EchoGenesis, an AI-based voice preservation system for individuals with progressive speech loss.
- Preserve a person's original voice using even short or partially degraded voice recordings.
- Generate clear, natural, and personalized speech that sounds like the user's own voice.
- Continuously adapt to changes in the user's voice caused by conditions such as ALS(Amyotrophic lateral sclerosis), laryngectomy, or dysarthria.
- Help users maintain their vocal identity and communicate effectively throughout the progression of their condition.

METHODOLOGY

EchoGenesis is a project that helps to save a person’s voice and use it again when they have difficulty speaking. First, the user’s healthy voice is recorded and stored. These voice samples are used as the main voice reference. Next, a speaker encoder studies the voice and finds the special features of the person’s voice. It can also work with unclear or low quality voice recordings. New voice recordings are compared with the old healthy recordings to keep the original voice identity. After that, a Few Shot Text to Speech model takes the user’s text and changes it into speech using the saved voice. For long sentences, the text is divided into small parts to make the speech clear and smooth. Finally, all the parts are joined to produce the user’s preserved voice. This helps the person communicate using a voice that sounds close to their original voice.

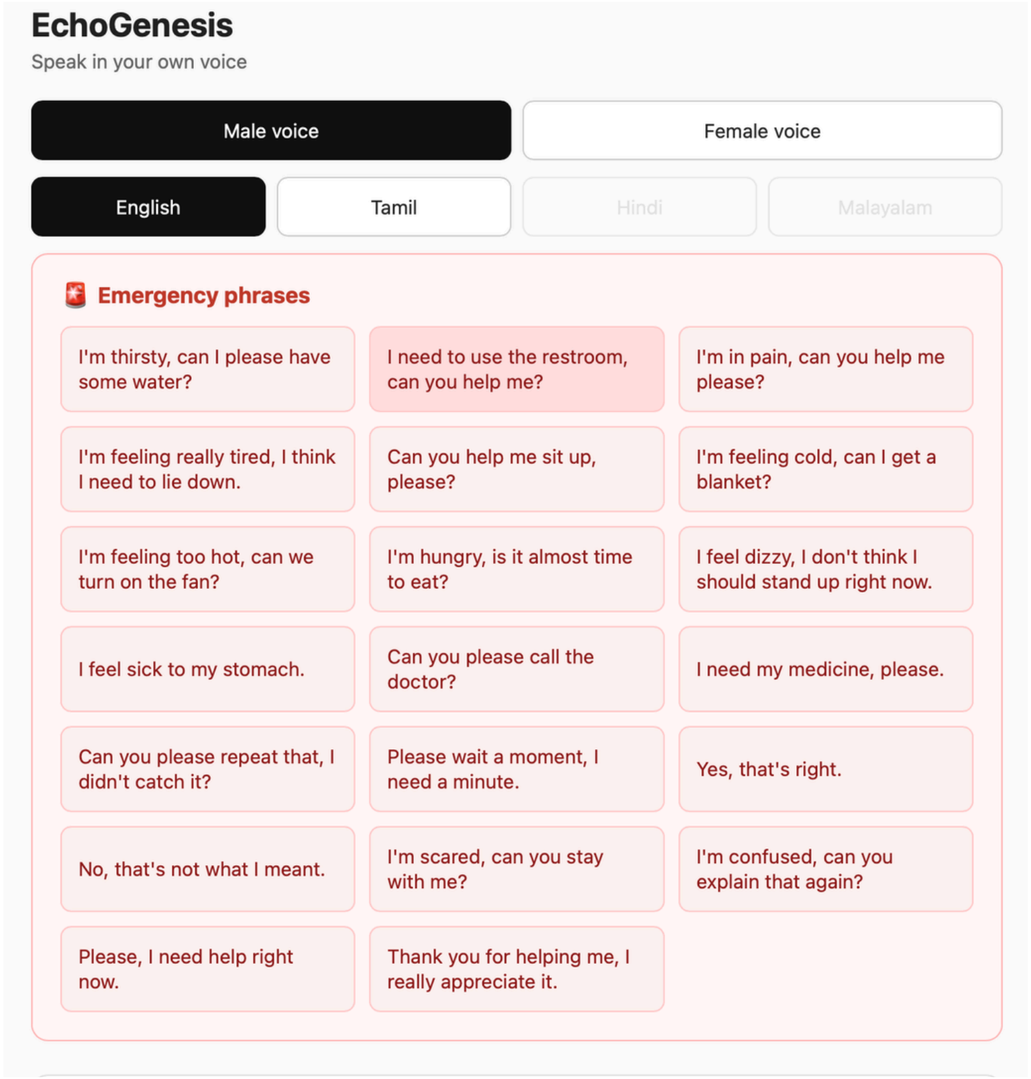
PIPELINE: VOICE BEYOND SILENCE - ECHOGENESIS

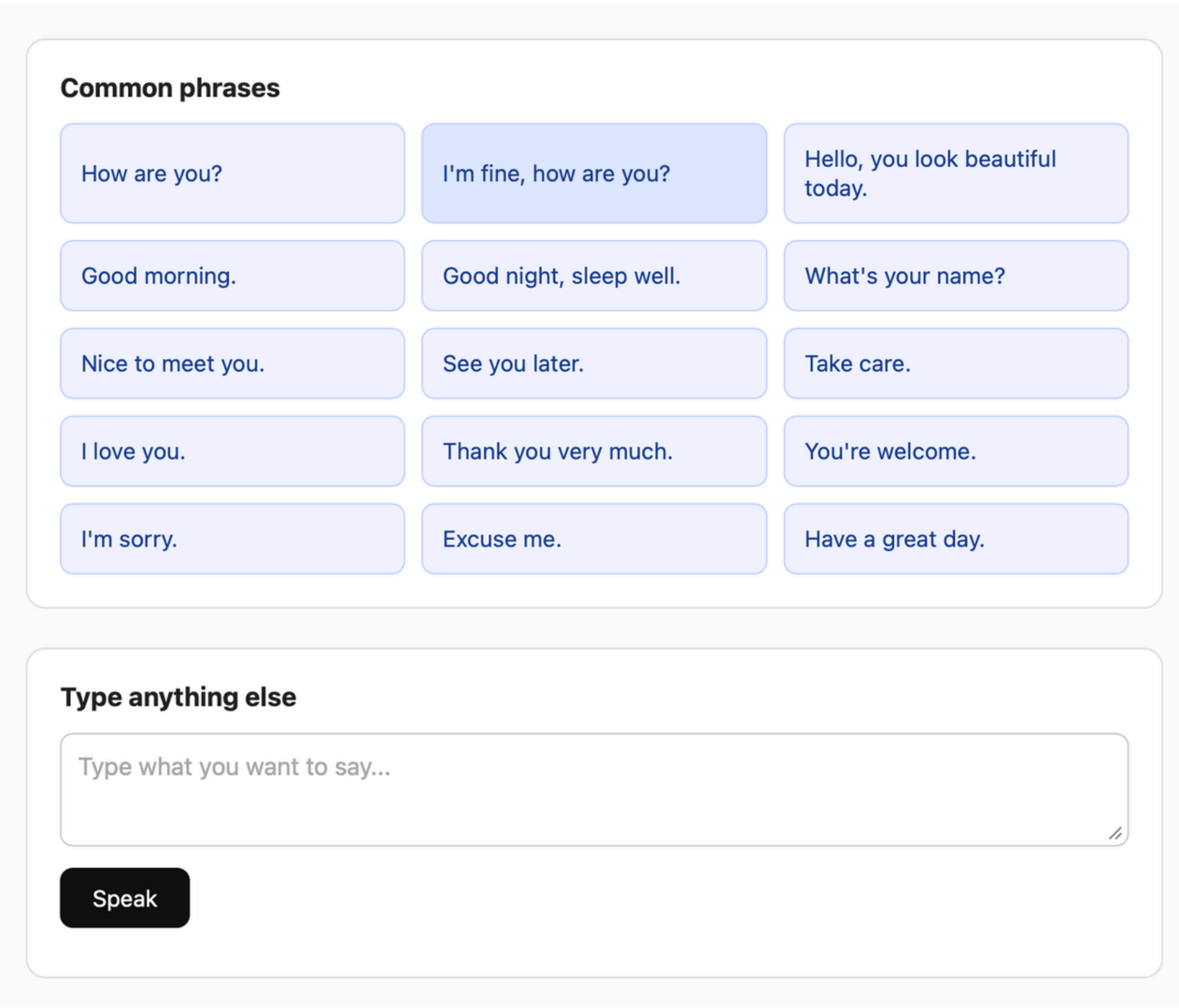


OUTCOME

The system takes old voice recordings, new voice recordings, and the text that needs to be spoken. It then generates clear and natural speech that sounds like the original voice, even when the speaker is no longer able to speak clearly. The generated speech should also sound smooth and natural, even for long sentences, without sounding robotic or broken. The system will be tested based on how closely the generated voice matches the original voice, how clearly and correctly the words are spoken, and how well the original voice is preserved even as the speech condition gets worse over time.

OUTCOME





- **Implementation of EchoGenesis**

TECHNOLOGY STACK

Component	Library/Model	Purpose
Voice cloning TTS	XTTS-v2 (Coqui TTS	Core speech synthesis engine, multilingual, supports reference-audio voice cloning
Fine-tuned checkpoint	finetuned_model/male/best_model.pth	Fine tuned on top of base XTTS-v2
Speech-to-text	Whisper (OpenAI, base model)	Transcribing recordings to verify/auto-match them against expected scripts
Speaker embeddings (legacy)	SpeechBrain spkrec-xvect-voxceleb	x-vector extraction, used in an earlier pipeline iteration
Noise reduction	noisereduce	Spectral-gating background noise removal on all recordings
Audio I/O/processing	librosa, soundfile, ffmpeg	Loading, resampling, trimming, format conversion (opus → wav)
Deep learning backend	PyTorch, torchaudio	Underlying XTTS/SpeechBrain inference

1. Voice Cloning Module

This module uses the XTTS-v2 model to generate speech that sounds similar to the original speaker. It takes a reference voice recording and the required text as input. The model generates speech while trying to preserve the speaker's voice identity.

Input: Reference voice recording and text.

Output: Speech generated in a similar voice.

2. Model Fine-Tuning Module

This module improves voice generation using a fine-tuned model checkpoint. It uses the trained model file, best_model.pth, along with the base XTTS-v2 model. The purpose is to improve the generated voice based on the training performed.

Input: Fine-tuned model checkpoint.

Output: An adapted voice-generation model.

3. Speech-to-Text Module

This module uses OpenAI Whisper to convert recorded speech into text. The recognized text can be compared with the expected script to check whether the recording matches the intended words.

Input: Voice recording.

Output: Text extracted from the recording.

4. Noise Reduction Module

This module uses the noisereduce library to reduce unwanted background sounds in voice recordings. Cleaner recordings help the system process speech more effectively.

Input: Noisy voice recording.

Output: A recording with reduced background noise.

5. Audio Processing Module

This module uses Librosa, SoundFile, and FFmpeg to prepare audio recordings for processing. It performs operations such as loading audio, trimming unwanted portions, changing the sample rate, and converting audio formats.

Input: Raw audio files.

Output: Processed audio suitable for the voice-generation system.

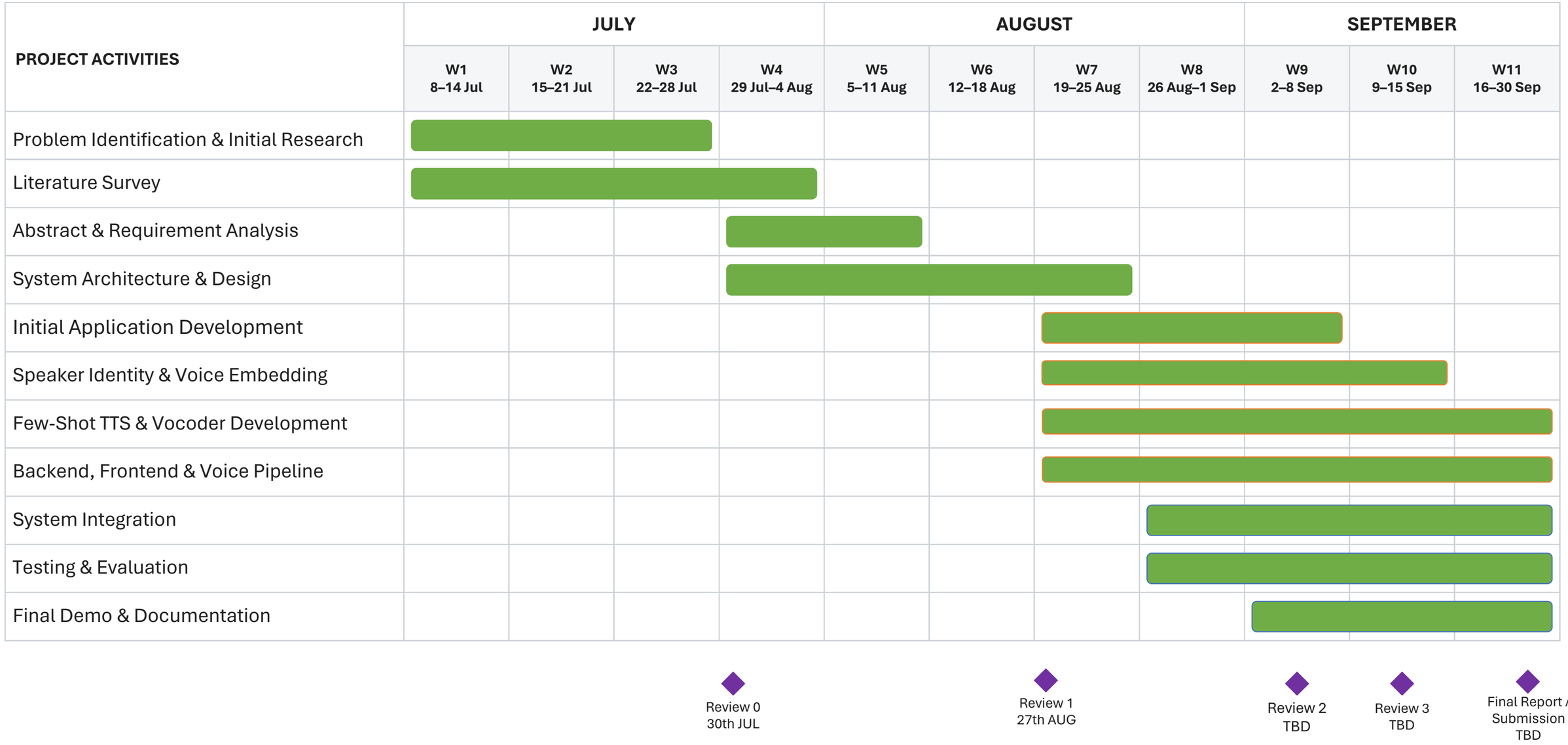
6. Deep Learning Backend Module

This module uses PyTorch and TorchAudio to support the deep-learning operations required by the system. These libraries help load models, process audio data, and execute model inference.

Input: Processed audio and model data.

Output: Results required for voice processing and speech generation.

ECHOGENESIS – PROJECT GANTT CHART



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